



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

 Website: www.ktunotes.in

4. What are the control strategies for the regulation of output voltage in voltage controllers?
5. A single-phase half bridge inverter has a resistive load of $10\ \Omega$, and input voltage of 96 V. Obtain the Fourier series representation of the waveform and hence find the value of distortion factor.
6. For a single-phase voltage controller, develop a relationship between conduction angle and firing angle. Under what condition conduction angle equals firing angle?
7. Design a simple light dimmer circuit using TRIAC including the triac firing circuit.
8. With the help of circuit diagram explain the working of current source inverter.
9. What is pulse width modulation? List the various PWM techniques.
10. Explain the principle of phase control in a single phase ac voltage controller.
11. Describe the working of a three phase voltage source inverter with a circuit diagram.
12. Write Fourier series expression for the output voltage from the single phase half bridge and full bridge inverter and determine the equation for THD.
13. Sketch the diagram and output voltage waveform of a single phase Voltage Source Inverter with R load and describe the working.
14. Define modulation index and Frequency modulation ratio.
15. What are the control strategies for the regulation of output voltage in Voltage Source Inverters?
16. Differentiate a Current source inverter from a Voltage source Inverter.
17. Describe the working of a three phase voltage source inverter with a circuit diagram.
18. Draw the phase and line voltage waveform of the three phase voltage source inverter with star connected resistive load on the assumption that each IGBT has a conduction angle of 120° .
19. A single phase bridge inverter fed from 200 V dc, is connected to an inductive load of $10\ \Omega$ and $L = 0.04\text{ H}$. Determine the power delivered to the load in case of continuous conduction mode operating at 50 Hz with square wave output.
20. With the help of waveform explain sinusoidal pulse width modulation in a three phase inverter.

Part B

1. a) Explain the 120° conduction mode of a three-phase bridge inverter, indicating the devices conducting in each state.
b) Write short notes of THD. (4)
2. a) Explain sinusoidal PWM technique for varying the magnitude of a single-phase inverter. (6)
b) Briefly explain current source inverter (8)
3. Explain the operation of 3 phase voltage source inverter with 180° conduction mode.
4. Compare voltage source and current source inverters.
5. What is sequence control in single phase ac voltage controllers? What are the advantages of employing it?
6. Explain sine PWM technique, with relevant waveforms. Define modulation index and mention its significance.
7. Explain the 120° conduction mode of a three-phase bridge inverter, indicating the devices conducting in each state.
8. Explain the operation of single phase voltage controller with RL load, drawing the voltage and current waveforms.
9. Draw the phase and line voltage waveform of the three phase voltage source inverter with star connected resistive load on the assumption that each IGBT conducts for 120°.
10. Draw the circuit and explain the 180° operation of a 3 phase bridge inverter with resistive load. Draw the phase voltage and line voltage waveforms.
11. Illustrate the operation of a 3-phase bridge inverter operating in 180° conduction mode with output line voltage and phase voltage waveforms. Derive the expression for output line voltage and phase voltage.
12. What is meant by Pulse Width Modulation? Describe the various PWM techniques for Voltage control of Inverters.